

Machine Learning 1

Mock Exam B – Answer Key

Worked solutions for all 100 points

Equivalent algebraic forms are valid. The derivations below use numerator-layout derivatives where a scalar derivative is a row vector.

Exercise 1: Normalization and a nonsmooth loss

1. Component derivative

Answer. Let $\rho = \|x\|_2 > 0$. Since

$$h_i(x) = x_i \rho^{-1}, \quad \frac{\partial \rho}{\partial x_j} = \frac{x_j}{\rho},$$

the product rule gives

$$\boxed{\frac{\partial h_i}{\partial x_j} = \frac{\delta_{ij}}{\rho} - \frac{x_i x_j}{\rho^3}}.$$

2. Full Jacobian

Answer. With $u = x/\|x\|_2$,

$$\boxed{J_h(x) = \frac{I_d}{\|x\|_2} - \frac{xx^\top}{\|x\|_2^3} = \frac{1}{\|x\|_2} (I_d - uu^\top)} \in \mathbb{R}^{d \times d}.$$

3. Radial direction

Answer.

$$J_h(x)x = \frac{x}{\rho} - \frac{x(x^\top x)}{\rho^3} = \frac{x}{\rho} - \frac{x\rho^2}{\rho^3} = 0.$$

For $c > 0$,

$$h(cx) = \frac{cx}{\|cx\|_2} = \frac{cx}{c\|x\|_2} = h(x).$$

Thus h is constant along each positive radial ray, so its directional derivative in the radial direction is zero.

4. Absolute-error derivative

Answer. Let $z = a^\top w - y$. For $z \neq 0$,

$$\frac{\partial |z|}{\partial w} = \text{sign}(z)a^\top, \quad \frac{\partial}{\partial w} \frac{\lambda}{2} w^\top w = \lambda w^\top.$$

Therefore

$$\boxed{\frac{\partial f}{\partial w} = \text{sign}(a^\top w - y)a^\top + \lambda w^\top} \in \mathbb{R}^{1 \times p}.$$

5. Nondifferentiability and subgradient

Answer. At $a^\top w - y = 0$, the absolute-value term has different one-sided slopes in direction a ; the smooth quadratic does not remove this kink. The row-subdifferential is

$$\partial f(w) = \{sa^\top + \lambda w^\top : s \in [-1, 1]\}.$$

For any chosen $s \in [-1, 1]$, one valid column-form update is

$$w^+ = w - \eta(sa + \lambda w).$$

For example, $s = 0$ gives $w^+ = (1 - \eta\lambda)w$ at the kink.

Exercise 2: A transformed waiting-time model

1. Valid density

Answer. The density is nonnegative. With $u = \lambda x^r$ and $du = r\lambda x^{r-1}dx$,

$$\int_{-\infty}^{\infty} p(x | \lambda) dx = \int_0^{\infty} r\lambda x^{r-1} e^{-\lambda x^r} dx = \int_0^{\infty} e^{-u} du = 1.$$

Thus it is a valid density for every $r, \lambda > 0$.

2. Distribution of $Y = X^r$

Answer. For $y > 0$, $x = y^{1/r}$ and

$$\left| \frac{dx}{dy} \right| = \frac{1}{r} y^{1/r-1}.$$

Hence

$$\begin{aligned} p_Y(y | \lambda) &= r\lambda (y^{1/r})^{r-1} e^{-\lambda y} \frac{1}{r} y^{1/r-1} \\ &= \lambda e^{-\lambda y}, \quad y > 0. \end{aligned}$$

Therefore

$$Y = X^r \mid \lambda \sim \text{Exponential}(\lambda), \quad \mathbb{E}[X^r \mid \lambda] = \frac{1}{\lambda}.$$

3. Survival probability

Answer.

$$P(X > c \mid \lambda) = P(X^r > c^r \mid \lambda) = e^{-\lambda c^r}.$$

For the supplied values,

$$P(X > 2 \mid \lambda = 0.25) = e^{-0.25 \cdot 2^2} = e^{-1}.$$

4. Maximum-likelihood estimator

Answer. Terms independent of λ can be collected into a constant:

$$L(\lambda) \propto \lambda^N e^{-\lambda S}, \quad \log L(\lambda) = N \log \lambda - \lambda S + \text{constant}.$$

Thus

$$\frac{d}{d\lambda} \log L = \frac{N}{\lambda} - S = 0 \implies \hat{\lambda}_{\text{ML}} = \frac{N}{S}.$$

The second derivative $-N/\lambda^2$ is negative. Since $S > 0$, the log-likelihood also tends to $-\infty$ at both boundaries, so this is the unique global maximum.

5. Posterior distribution

Answer.

$$\begin{aligned} p(\lambda \mid x_1, \dots, x_N) &\propto \lambda^N e^{-S\lambda} \lambda^{a-1} e^{-b\lambda} \\ &= \lambda^{a+N-1} e^{-(b+S)\lambda}. \end{aligned}$$

Therefore, in shape-rate form,

$$\lambda \mid x_1, \dots, x_N \sim \text{Gamma}(a + N, b + S).$$

6. MAP estimator

Answer. The derivative of the posterior log-density is

$$\frac{a + N - 1}{\lambda} - (b + S).$$

Since $N \geq 1$ and $a > 0$, the posterior shape is greater than one. Hence the unique interior mode is

$$\hat{\lambda}_{\text{MAP}} = \frac{a + N - 1}{b + S}.$$

7. Posterior mean and asymptotics

Answer.

$$\mathbb{E}[\lambda \mid \text{data}] = \frac{a + N}{b + S}, \quad \mathbb{E}[\lambda \mid \text{data}] - \hat{\lambda}_{\text{MAP}} = \frac{1}{b + S}.$$

If $S/N \rightarrow s > 0$, the difference is $1/(b + Ns) \rightarrow 0$ at rate $O(N^{-1})$. Both summaries converge to $1/s$.

Exercise 3: Bias, variance, and shrinkage

Write $A = \sigma^2/N > 0$ and $D = (\mu - m_0)^2 \geq 0$.

1. Bias

Answer.

$$\mathbb{E}[\tilde{\mu}_c] = c\mu + (1 - c)m_0,$$

so

$$\text{Bias}(\tilde{\mu}_c) = (1 - c)(m_0 - \mu).$$

The estimator is unbiased exactly when $c = 1$ or $m_0 = \mu$.

2. Variance

Answer. Independence gives $\text{Var}(\bar{X}) = \sigma^2/N$. Since m_0 is fixed,

$$\text{Var}(\tilde{\mu}_c) = \frac{c^2 \sigma^2}{N}.$$

3. Mean squared error

Answer.

$$\text{MSE}(c) = \frac{c^2 \sigma^2}{N} + (1 - c)^2 (m_0 - \mu)^2 = Ac^2 + D(1 - c)^2.$$

4. MSE-minimizing value

Answer. Differentiation gives

$$\frac{d}{dc} \text{MSE}(c) = 2(A + D)c - 2D.$$

Therefore

$$c^* = \frac{D}{A + D} = \frac{(\mu - m_0)^2}{(\mu - m_0)^2 + \sigma^2/N}.$$

This lies in $[0, 1)$, and the second derivative $2(A + D) > 0$, so it is the unique minimizer.

5. Oracle property and limit

Answer. This is an oracle choice because it depends on the unknown μ . If $\mu \neq m_0$, then $D > 0$ and $c^* \rightarrow 1$ as $N \rightarrow \infty$, so the estimator increasingly relies on the precise sample mean. If $\mu = m_0$, then $D = 0$ and $c^* = 0$ for every N ; the fixed reference is exactly correct and has zero MSE.

Exercise 4: Multi-output MAP regression

Define

$$D(W) = \|T - \Phi W\|_F^2, \quad P(W) = \text{tr}((W - M_0)^\top R(W - M_0)).$$

1. Negative log-posterior

Answer. Up to constants independent of W ,

$$-\log p(W \mid T) = \frac{\beta}{2} D(W) + \frac{\alpha}{2} P(W) + \text{constant}.$$

2. MAP estimator

Answer. Differentiation with respect to W gives the stationarity equation

$$\beta \Phi^\top (\Phi W - T) + \alpha R(W - M_0) = 0.$$

With $\lambda = \alpha/\beta$,

$$(\Phi^\top \Phi + \lambda R)W = \Phi^\top T + \lambda R M_0.$$

Therefore

$$W_{\text{MAP}} = (\Phi^\top \Phi + \lambda R)^{-1} (\Phi^\top T + \lambda R M_0).$$

An equivalent form is

$$W_{\text{MAP}} = M_0 + (\Phi^\top \Phi + \lambda R)^{-1} \Phi^\top (T - \Phi M_0).$$

3. Uniqueness

Answer. For every nonzero $v \in \mathbb{R}^m$,

$$v^\top (\Phi^\top \Phi + \lambda R) v = \|\Phi v\|_2^2 + \lambda v^\top R v > 0.$$

Thus the coefficient matrix is positive definite and invertible, regardless of the rank of Φ . The objective is strictly convex, so the MAP estimate is unique.

4. Limits

Answer. If Φ has full column rank,

$$\lim_{\lambda \rightarrow 0} W_{\text{MAP}} = (\Phi^\top \Phi)^{-1} \Phi^\top T,$$

the ordinary least-squares estimate. Using the equivalent form above,

$$\lim_{\lambda \rightarrow \infty} W_{\text{MAP}} = M_0,$$

because the correction away from M_0 is $O(\lambda^{-1})$.

If Φ is rank deficient, the first displayed inverse does not exist; the small- λ limit selects, among least-squares solutions, the one closest to M_0 in the R -weighted norm.

5. Regularization path and bias–variance tradeoff

Answer. Let W_i minimize $D(W) + \lambda_i P(W)$ for $0 < \lambda_1 < \lambda_2$, and write $D_i = D(W_i)$ and $P_i = P(W_i)$. Optimality gives

$$\begin{aligned} D_1 + \lambda_1 P_1 &\leq D_2 + \lambda_1 P_2, \\ D_2 + \lambda_2 P_2 &\leq D_1 + \lambda_2 P_1. \end{aligned}$$

Combining these inequalities shows

$$P_2 \leq P_1, \quad D_2 \geq D_1.$$

Thus stronger regularization moves the estimate toward M_0 and cannot improve the training residual; the inequalities need not be strict.

Under a correctly specified fixed-design model $T = \Phi W_* + E$ with mean-zero isotropic noise, put $A_\lambda = \Phi^\top \Phi + \lambda R$. Then

$$\mathbb{E}[W_{\text{MAP}}] - W_* = \lambda A_\lambda^{-1} R (M_0 - W_*).$$

Stronger shrinkage generally reduces variance but increases bias toward M_0 . If $M_0 = W_*$, the bias is zero for every λ , while the variance still decreases.

Exercise 5: Two sequential Bayesian updates

Let $\Lambda_N = \Sigma_N^{-1}$.

1. Final precision

Answer. Each Gaussian observation adds a rank-one precision term. Hence

$$\Lambda_{\text{final}} = \Lambda_N + \beta_a \phi_a \phi_a^\top + \beta_b \phi_b \phi_b^\top,$$

and $\Sigma_{\text{final}} = \Lambda_{\text{final}}^{-1}$.

2. Natural parameter and posterior mean

Answer. The linear terms in w give

$$\eta_{\text{final}} = \eta_N + \beta_a \phi_a t_a + \beta_b \phi_b t_b.$$

Therefore

$$\mu_{\text{final}} = \Sigma_{\text{final}} \eta_{\text{final}}$$

or, explicitly,

$$\mu_{\text{final}} = (\Sigma_N^{-1} + \beta_a \phi_a \phi_a^\top + \beta_b \phi_b \phi_b^\top)^{-1} (\Sigma_N^{-1} \mu_N + \beta_a \phi_a t_a + \beta_b \phi_b t_b).$$

3. Order independence

Answer. Processing either observation simply adds its precision and natural-parameter terms. Matrix and vector addition are commutative, so both processing orders produce the same final Gaussian posterior. This requires conditional independence:

$$p(t_a, t_b \mid w, \phi_a, \phi_b) = p(t_a \mid w, \phi_a) p(t_b \mid w, \phi_b).$$

Equivalently, the two observation noises must be independent conditional on w and the features.

4. Equivalent observation for equal features

Answer. If $\phi_b = \phi_a = \phi$, then the increments are

$$\Delta\Lambda = (\beta_a + \beta_b) \phi \phi^\top, \quad \Delta\eta = \phi(\beta_a t_a + \beta_b t_b).$$

Define

$$\beta_* = \beta_a + \beta_b, \quad t_* = \frac{\beta_a t_a + \beta_b t_b}{\beta_a + \beta_b}.$$

Then $\Delta\Lambda = \beta_* \phi \phi^\top$ and $\Delta\eta = \beta_* \phi t_*$, exactly the update from one observation (ϕ, t_*, β_*) .

5. Collinear features

Answer. If $\phi_b = c\phi_a$, then

$$\beta_b \phi_b \phi_b^\top = \beta_b c^2 \phi_a \phi_a^\top.$$

Thus the second observation adds precision only in $\text{span}(\phi_a)$. It provides more information in the same direction, but no new parameter-space direction. If $\phi_a = 0$, neither observation gives information about w .